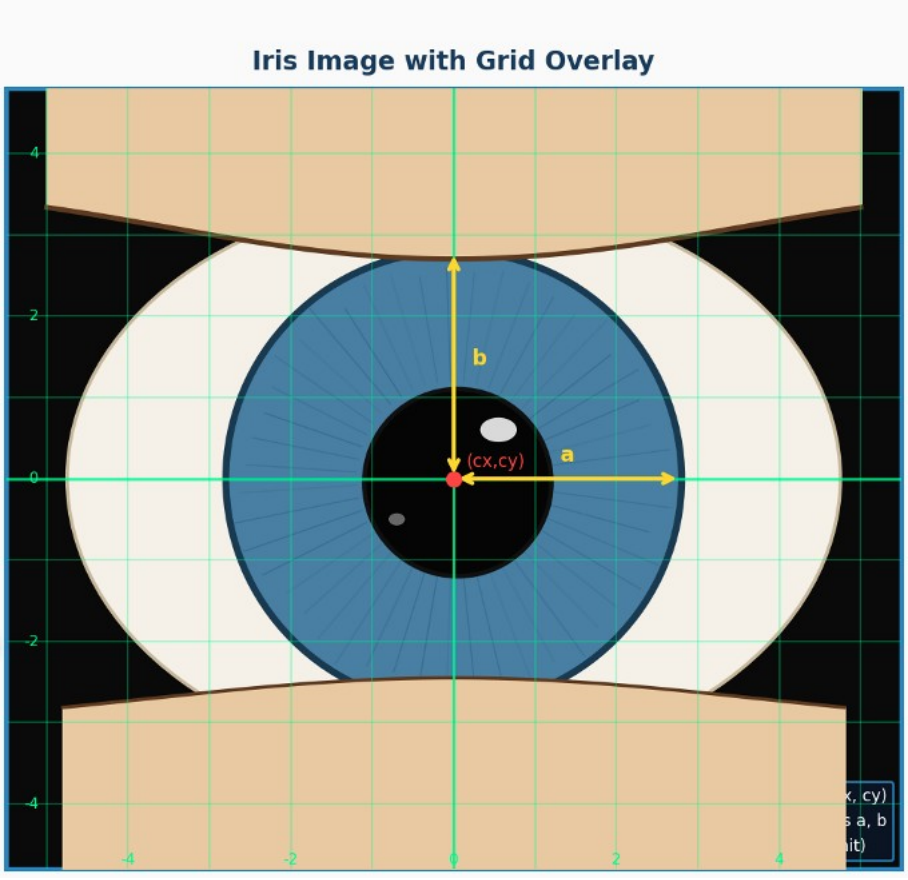
Unit 3 - Session 5

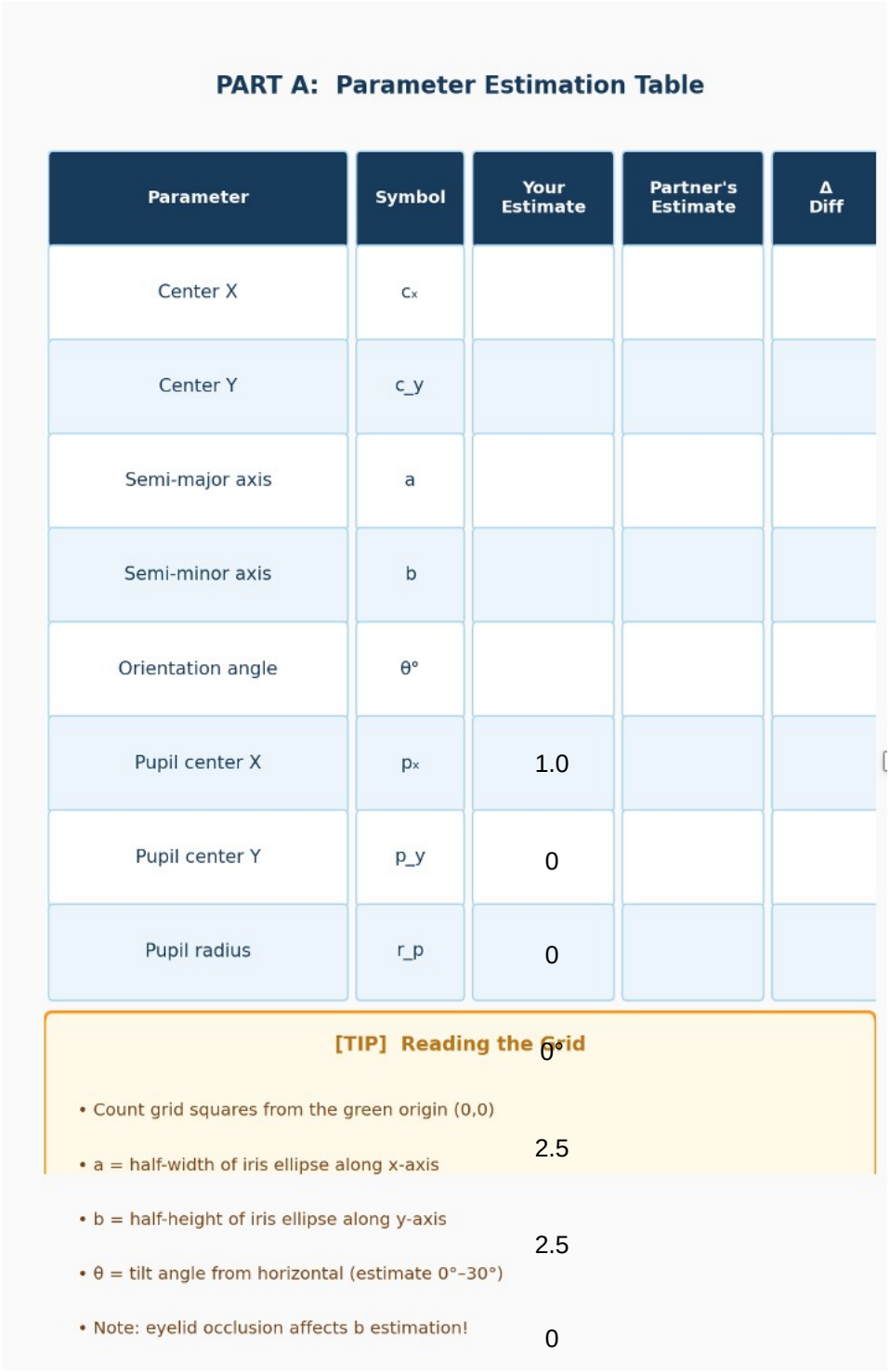
SLO 2: Adapt HT constraints for biometric applications

Duration: 10 Minutes Activity

Question:

Answer:
Center X (c_x) = 0
Center Y (c_y) = 0
Semi-major axis (a) ≈ 2.5
Semi-minor axis (b) ≈ 2.5
Orientation angle (θ°) ≈ 0°
Pupil center X (p_x) = 0
Pupil center Y (p_y) = 0
Pupil radius (r_p) ≈ 1.0





PART A: Parameter Estimation Table

Parameter	Symbol	Your Estimate	Partner's Estimate	Δ Diff
Center X	c_x			
Center Y	c_y			
Semi-major axis	a			
Semi-minor axis	b			
Orientation angle	θ°			
Pupil center X	p_x	1.0		
Pupil center Y	p_y	0		
Pupil radius	r_p	0		

[TIP] Reading the Grid

- Count grid squares from the green origin (0,0)
- a = half-width of iris ellipse along x-axis
- b = half-height of iris ellipse along y-axis
- θ = tilt angle from horizontal (estimate 0°–30°)
- Note: eyelid occlusion affects b estimation!

0°

2.5

2.5

0

0

"Spot the Ellipse" Parameter Hunt. Then they compare their estimates in pairs and discuss how lighting/occlusion changes the parameters. Reinforces why robust HT voting matters in noisy biometric data.